

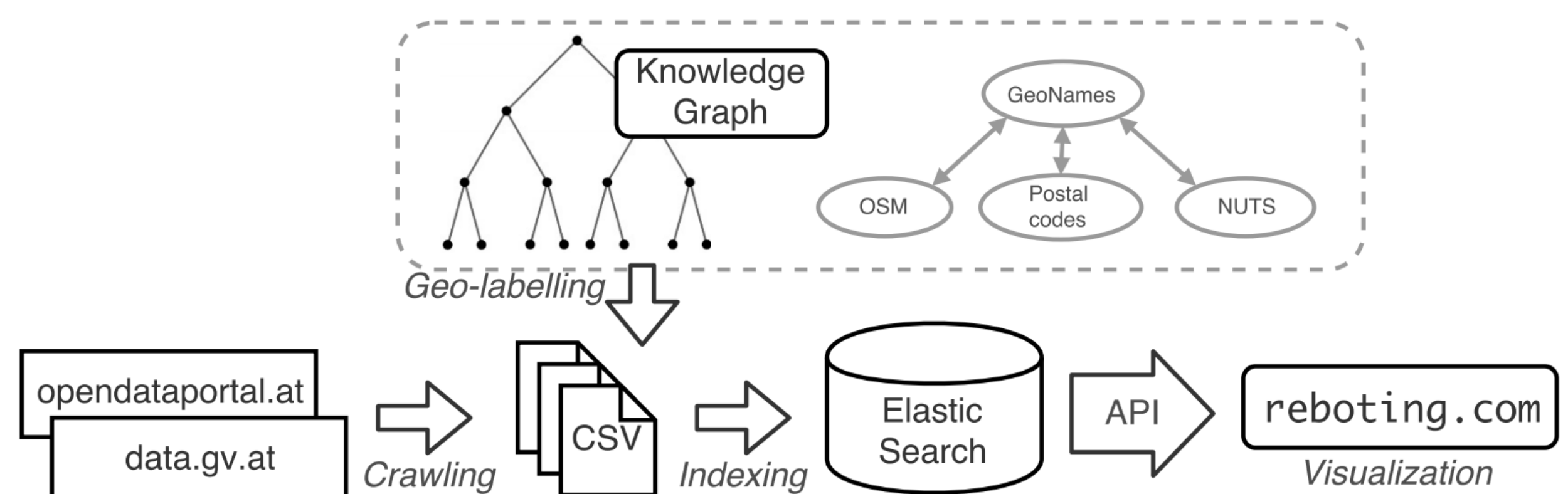
INTRODUCTION

Data portals mainly publish semi-structured, tabular formats which lack semantic descriptions of geo-entities and therefore, do not allow any exploration and automated visualization of these datasets.

We present a framework to add geo-semantic labels, based on a constructed geo-entity knowledge graph, and a user interface to query and automatically visualize the resources from the Austrian data portals.

APPROACH

1. Crawl CSVs and metadata from the Austrian data portals
2. Label datasets using our constructed base knowledge graph of geo-entities
3. Store and index datasets in a search engine
4. Automatically generate (geo-) visualizations



GEO-SEMANTIC LABELLING & VISUALIZATION

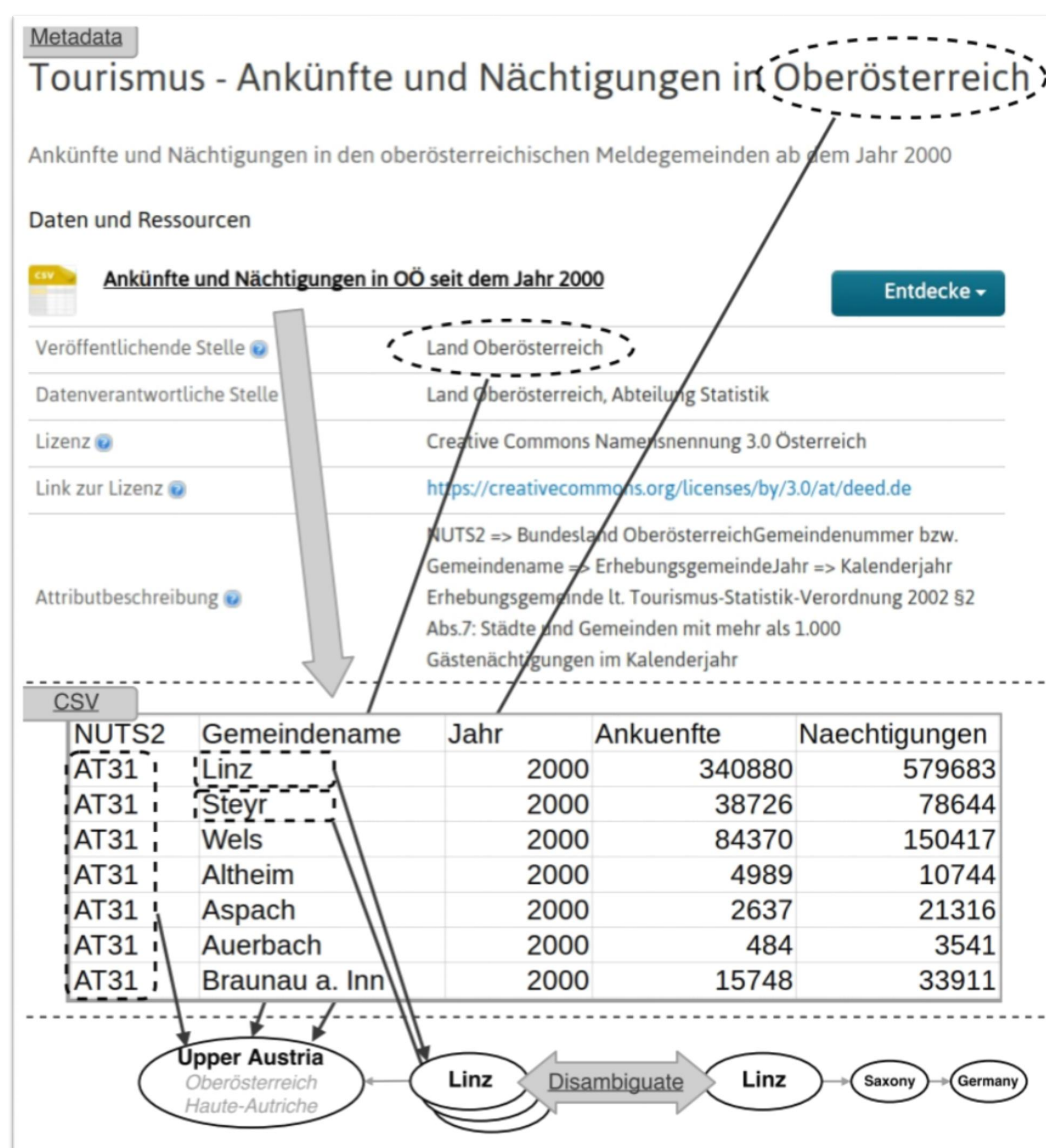


Fig. 1: Extraction and disambiguation of geo-information based on the hierarchical base knowledge graph

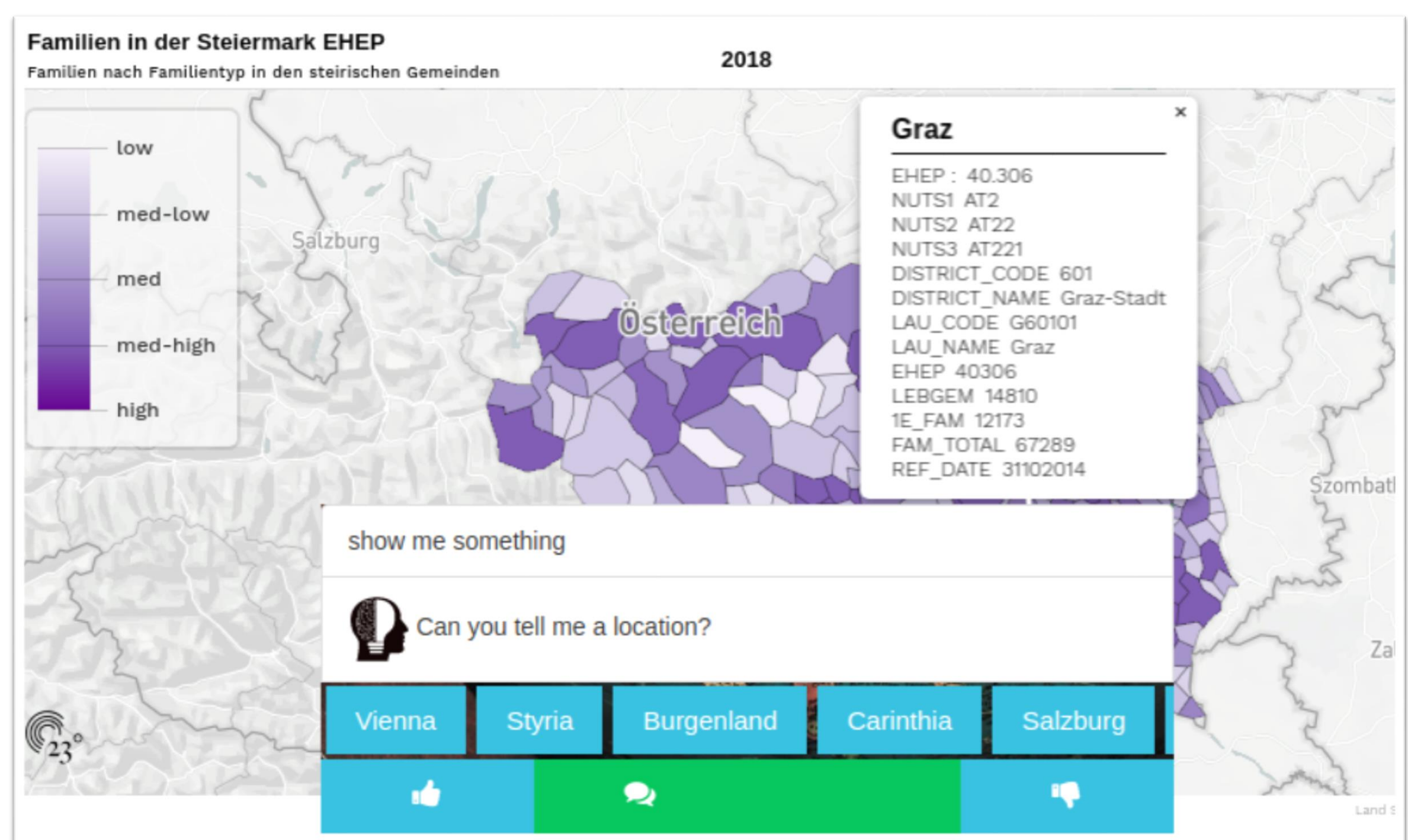


Fig. 2: Automatically generated visualizations for the federal states of Austria

FUTURE WORK

- Use user ratings to adapt & select visualizations
- Scale system to datasets/portals worldwide
- Make base knowledge graph and annotated CSVs via GeoSPARQL endpoint available

GEO-ENTITY KNOWLEDGE GRAPH

We construct a KG by integrating GeoNames, OSM, NUTS and national postal codes.



Find more information here:
<http://data.wu.ac.at/odgraphsearch/>

REFERENCES

- Portal Watch* Neumaier, S., Umbrich, J., Polleres, A.: Automated Quality Assessment of Metadata across Open Data Portals. ACM J. Data Inform. Quality, 2016.
- GeoSPARQL* Perry, M., Herring, J.: OGC GeoSPARQL - A geographic query language for RDF data. OGC Implementation Standard. 2012